1) Fact_Trip Analysis

📌 Represents

One completed taxi trip transaction.

---

📊 Possible Insights

- Total trips
- Average trip duration
- Average trip distance
- Passenger behavior
- Peak travel periods
- Most active pickup locations
- Trips by date
- Trips by vendor

---

1. Total Trips

📊 Insight

Shows the total number of completed taxi trips.

```
SELECT COUNT(*) AS TotalTrips
FROM Fact_Trip;
```

---

2. Average Trip Duration

📊 Insight

Measures the average duration of taxi trips.

```
SELECT AVG(trip_duration_sec) AS AvgTripDuration
FROM Fact_Trip;
```

---

3. Average Trip Distance

📊 Insight

Shows the average travel distance per trip.

```
SELECT AVG(trip_distance_km) AS AvgTripDistance
FROM Fact_Trip;
```

---

4. Passenger Behavior

📊 Insight

Analyzes how many passengers usually travel in one trip.

```
SELECT AVG(passenger_count) AS AvgPassengers
FROM Fact_Trip;
```

---

5. Peak Travel Periods

📊 Insight

Identifies the busiest hours for taxi trips.

```sql
SELECT DT.hour,
    COUNT(*) AS TotalTrips
FROM Fact_Trip FT
JOIN Dim_Time DT
ON FT.time_key = DT.time_key
GROUP BY DT.hour
ORDER BY TotalTrips DESC;
```

---

### 6. Most Active Pickup Locations

📊 Insight

Shows which pickup locations generate the highest number of trips.

```sql
SELECT DL.location_name,
    COUNT(*) AS TotalTrips
FROM Fact_Trip FT
JOIN Dim_Location DL
ON FT.pickup_location_key = DL.location_key
GROUP BY DL.location_name
ORDER BY TotalTrips DESC;
```

---

### 7. Trips by Vendor

📊 Insight

Compares trip counts between taxi vendors.

```sql
SELECT DV.vendor_name,
    COUNT(*) AS TotalTrips
FROM Fact_Trip FT
JOIN Dim_Vendor DV
ON FT.vendor_key = DV.vendor_key
GROUP BY DV.vendor_name
ORDER BY TotalTrips DESC;
```

---

### 8. Trips by Date

📊 Insight

Analyzes daily trip activity trends.

```sql
SELECT DD.full_date,
    COUNT(*) AS TotalTrips
FROM Fact_Trip FT
JOIN Dim_Date DD
ON FT.date_key = DD.date_key
GROUP BY DD.full_date
ORDER BY DD.full_date;
```

---

---

2) Fact_VendorPerformance Analysis

📌 Represents

Daily aggregated vendor performance.

---

📊 Possible Insights

- Best performing vendor
- Daily vendor activity
- Vendor operational trends
- Vendor passenger handling efficiency

---

9. Best Performing Vendor

📊 Insight

Determines which vendor completed the highest number of trips.

```
SELECT DV.vendor_name,
    SUM(FVP.total_trips) AS TotalTrips
FROM Fact_VendorPerformance FVP
JOIN Dim_Vendor DV
ON FVP.vendor_key = DV.vendor_key
GROUP BY DV.vendor_name
ORDER BY TotalTrips DESC;
```

---

10. Daily Vendor Activity

📊 Insight

Tracks vendor trip activity on each day.

```
SELECT DD.full_date,
    DV.vendor_name,
    FVP.total_trips
FROM Fact_VendorPerformance FVP
JOIN Dim_Date DD
ON FVP.date_key = DD.date_key
JOIN Dim_Vendor DV
ON FVP.vendor_key = DV.vendor_key
ORDER BY DD.full_date;
```

---

11. Vendor Operational Trends

📊 Insight

Analyzes average trip duration trends for each vendor.

```
SELECT DV.vendor_name,
    AVG(FVP.avg_trip_duration_sec) AS AvgTripDuration
FROM Fact_VendorPerformance FVP
JOIN Dim_Vendor DV
ON FVP.vendor_key = DV.vendor_key
```

```
GROUP BY DV.vendor_name;
```

---

12. Vendor Passenger Handling Efficiency

📊 Insight

Shows how efficiently vendors handle passenger loads.

```
SELECT DV.vendor_name,
    AVG(FVP.avg_passenger_count) AS AvgPassengerCount
FROM Fact_VendorPerformance FVP
JOIN Dim_Vendor DV
ON FVP.vendor_key = DV.vendor_key
GROUP BY DV.vendor_name;
```

---

---

3) Fact_HourlyDemand Analysis

📌 Represents

Taxi demand activity for each hour.

---

📊 Possible Insights

- Rush hours
- Peak demand periods
- Passenger density
- Hourly traffic patterns

---

13. Rush Hours

📊 Insight

Identifies hours with the highest number of trips.

```
SELECT DT.hour,
    SUM(FHD.trip_count) AS TotalTrips
FROM Fact_HourlyDemand FHD
JOIN Dim_Time DT
ON FHD.time_key = DT.time_key
GROUP BY DT.hour
ORDER BY TotalTrips DESC;
```

---

14. Peak Demand Periods

📊 Insight

Finds the top busiest hours of the day.

```
SELECT TOP 5
    DT.hour,
    SUM(FHD.trip_count) AS TotalTrips
FROM Fact_HourlyDemand FHD
JOIN Dim_Time DT
```

**ON FHD.time_key = DT.time_key**
**GROUP BY DT.hour**
**ORDER BY TotalTrips DESC;**

---

**15. Passenger Density**

📊 **Insight**

**Measures the total number of passengers during each hour.**

```
SELECT DT.hour,
    SUM(FHD.total_passengers) AS TotalPassengers
FROM Fact_HourlyDemand FHD
JOIN Dim_Time DT
ON FHD.time_key = DT.time_key
GROUP BY DT.hour
ORDER BY TotalPassengers DESC;
```

---

**16. Hourly Traffic Patterns**

📊 **Insight**

**Analyzes how trip durations change throughout the day.**

```
SELECT DT.hour,
    AVG(FHD.avg_duration_sec) AS AvgTripDuration
FROM Fact_HourlyDemand FHD
JOIN Dim_Time DT
ON FHD.time_key = DT.time_key
GROUP BY DT.hour
ORDER BY DT.hour;
```